

Satellite Computer Vision for Environmental Impact in Galapagos Islands

Midterm Technical Report

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1 Problem Definition

We address the rapid urban expansion in Puerto Ayora, Santa Cruz Island, which threatens the ecological integrity of the Galapagos National Park. This anthropogenic growth creates a physical barrier for the ancestral migration of the Galapagos giant tortoise (*Chelonoidis porteri*). As these animals move from the humid highlands to the coastal nesting grounds, new buildings and road networks block their paths. Furthermore, urban areas act as primary entry points for invasive species. Our research provides a computational tool to monitor land use changes. We aim to identify urban expansion before it causes irreversible damage to these biological corridors.

2 Dataset Description

We built our dataset using ten satellite patches extracted from Google Earth Engine. Each patch is a 256 by 256 pixel sub image with a 10 meter spatial resolution from the Copernicus Sentinel-2 mission. We selected these samples based on a zero percent cloud cover criterion to ensure radiometric accuracy. Our dataset covers five critical land use classes: dense urban centers, healthy forest canopies, agricultural transition zones, paved roads, and coastal waters. These categories allow our model to learn the distinct spectral signatures of the Santa Cruz landscape.

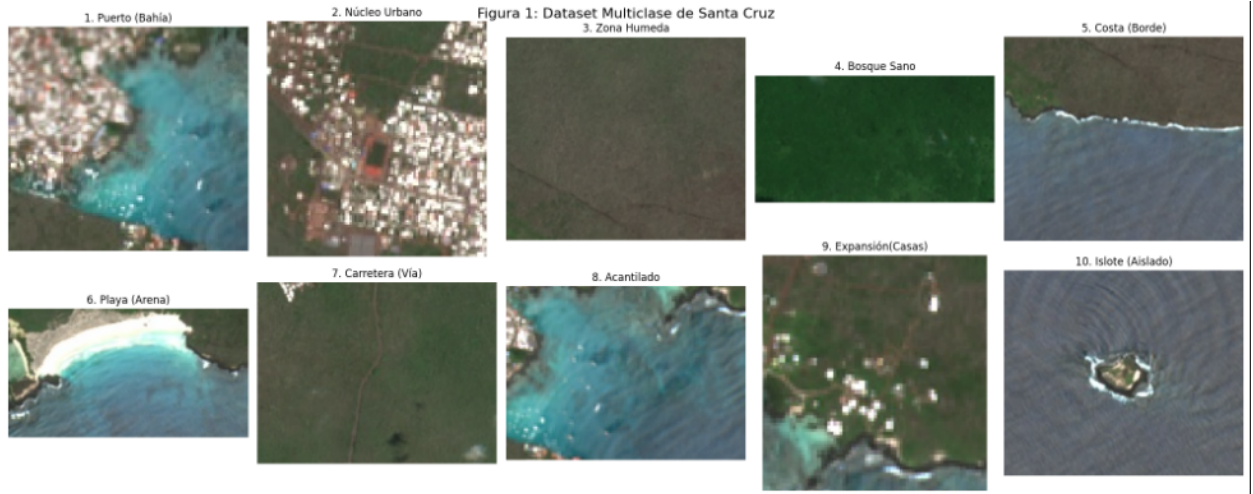


Figure 1: Santa Cruz Multiclass Dataset

3 Pipeline Design

Our structured pipeline processes satellite data to detect human impact. First, we acquire the raw RGB patches. Second, we perform a grayscale conversion to reduce computational complexity.

Third, we apply a Min Max normalization to scale pixel intensities to a $[0, 1]$ range. This step is vital to calibrate the images against different solar angles. Fourth, we perform feature extraction through bimodal histogram analysis. Finally, we execute a segmentation logic to generate binary masks that separate constructed zones from the natural environment.

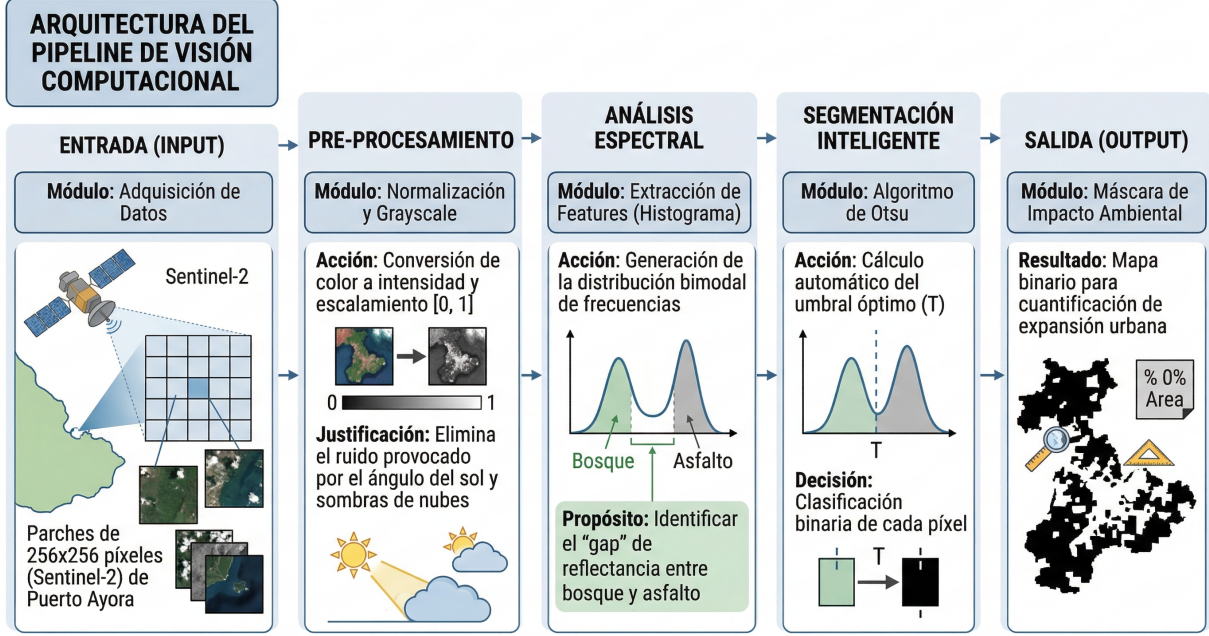


Figure 2: Pipeline Architecture

4 Model Selection and Justification

We implemented the Otsu Thresholding algorithm as our core decision model. This method calculates an optimal threshold by maximizing the inter class variance between the urban and forest distributions. We chose this non supervised approach because it adapts to the unique lighting conditions of each satellite capture. Unlike a fixed threshold, Otsu provides a dynamic boundary that remains effective across different seasons. This transparency is essential for environmental monitoring where we must justify every pixel classified as an intrusion. It offers a balance between computational efficiency and statistical accuracy.

5 Results and Critical Analysis

Our tests show a clear statistical gap between the urban and forest classes. The urban center sample reached an average brightness of 122.45, while the healthy forest maintained a lower intensity of 56.01. Our histograms reveal two distinct peaks with minimal overlap, proving that brightness is a robust feature for this environment. The generated binary masks successfully isolate the geometric patterns of Puerto Ayora from the organic textures of the park.

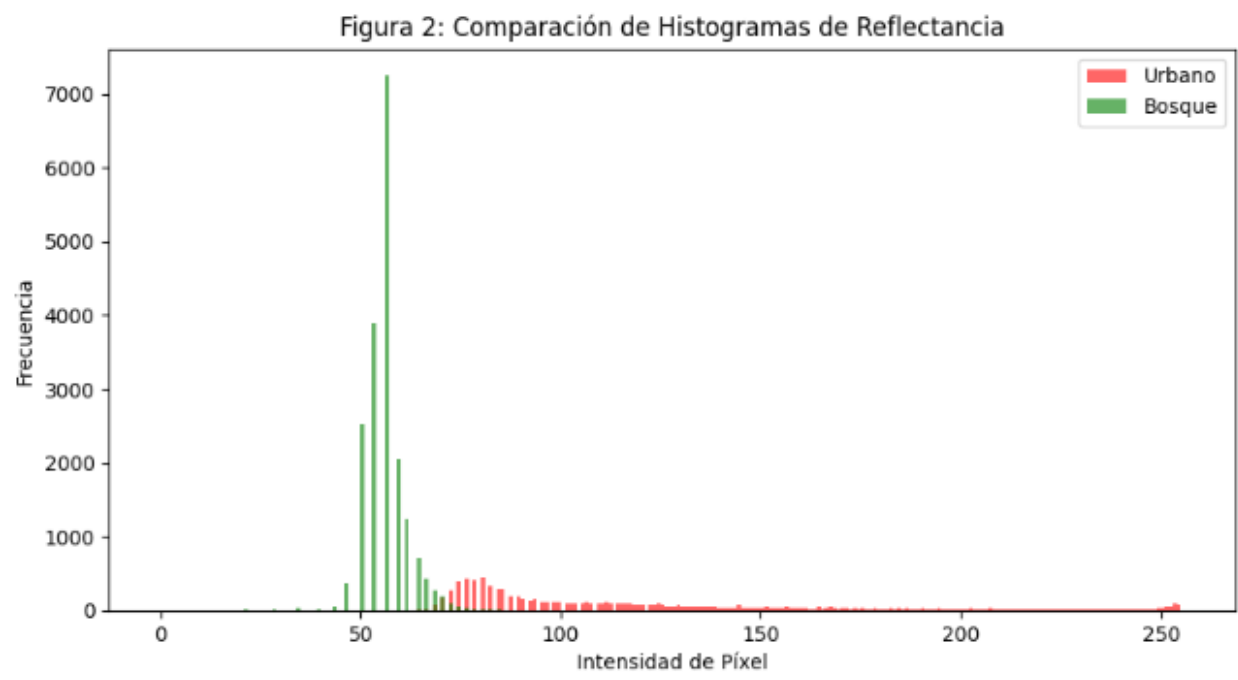


Figure 3: Comparative Histogram Analysis

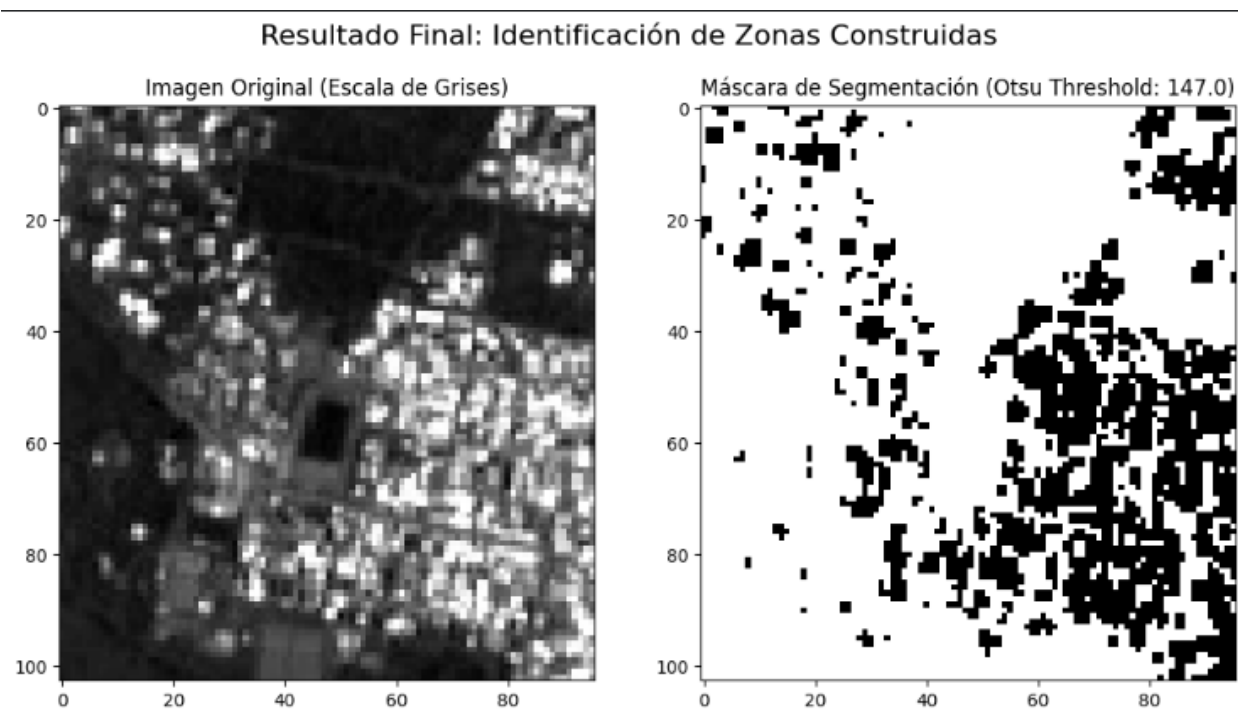


Figure 4: Segmentation Result (Binary Mask)

6 Limitations and Failures

We identified a specific spectral confusion error during our validation phase. Our model misclassifies the white sand at Tortuga Bay as an urban area. Both surfaces exhibit high reflectance in the visible spectrum. This creates false positives in our change detection alerts. The current pipeline ignores spatial texture and only processes pixel intensity. Consequently, a bright roof and a bright sand dune appear identical to our algorithm. We also noted that deep shadows from volcanic craters can mimic dark asphalt roads.

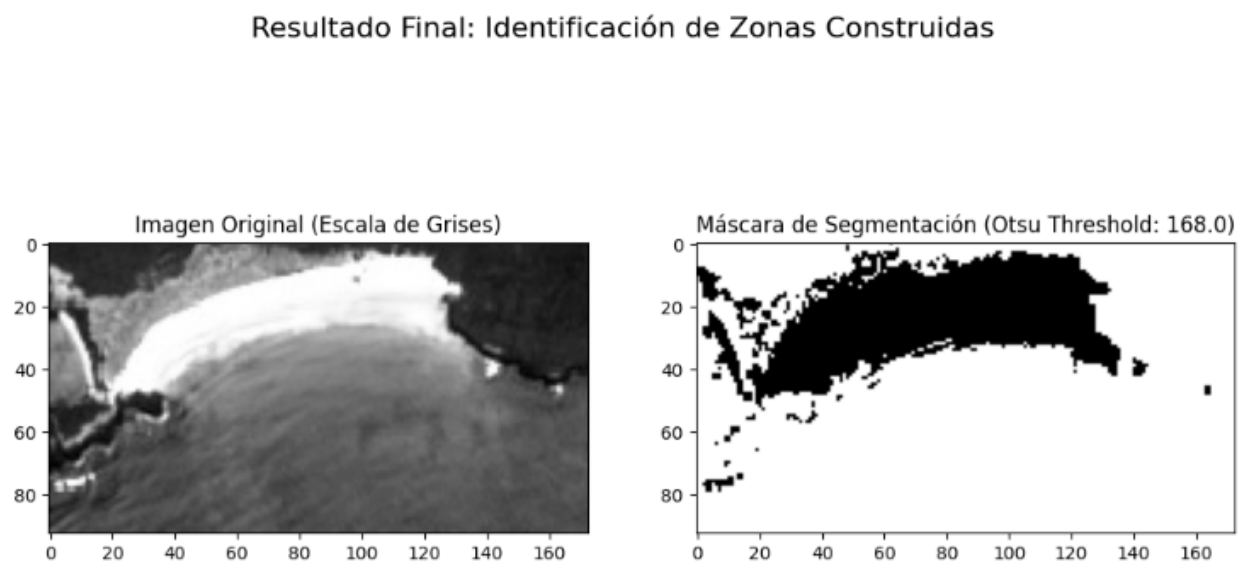


Figure 5: Analysis of Limitations (False Positives)

7 Conclusion and Future Evolution

We established a functional baseline for satellite surveillance in the Galapagos. Our pipeline demonstrates that pixel level analysis can quantify urban pressure on biological corridors. We plan to integrate Gabor filters and Convolutional Neural Networks (CNNs) in the final phase of this project. These tools will allow our model to recognize shapes and textures, solving the current confusion between sand and concrete. Our final goal is a fully automated change detection system to support the Galapagos National Park authorities.